Ayush Agrawal

ayushagrawal26@g.ucla.edu | +1 424-430-5322 | ayushagrawal149.github.io | https://www.linkedin.com/in/ayush145agrawal/

Education

University of California, Los Angeles (UCLA) — Los Angeles, CA

M.S. in Electrical and Computer Engineering, specializing in Robotics and Controls; ***GPA: 3.97/4.0*** — *June 2026*

- Awarded merit-based scholarship worth **$76,000** from JN Tata Endowment Fund and Narottam Sekhsaria Foundation
- Thesis: "*WorldBench: Benchmarking Physical Understanding of World Models By Isolating Physical Concepts*"

Indian Institute of Technology (IIT) Bombay — Mumbai, India

B.Tech. in Mechanical Engineering with a minor in Systems & Controls Engineering; ***GPA: 9.37/10.0*** — *June 2021*

- First-author oral presentations to 50+ researchers representing 40+ countries at IEEE CDC Singapore, and SAE SIAT India

Technical Skills

Programming	C++, Python, Pytorch, Model-in-loop testing (MiL), Software-in-loop testing (SiL), Linux
Robotics	ROS 1/2, Gazebo, Embedded C++, A*, RRT, Git, MISRA, ISO26262, ASIL-D, SolidWorks
Software	MATLAB/Simulink (6+ Years Experience), CAN protocol, PID, Jira, Confluence, AutoCAD
Relevant Coursework	Computer Vision (**A**), Deep Learning (**A**), Digital Control (**A+**), MPC - Motion Planning (**A**)
Professional Skills	Vehicle Dynamics, Systems Engineering, Safety Critical Control, Verification and Validation, CI/CD Workflows, Classical Control Theory, Root Cause Analysis, First Principles Modelling

Professional Experience

Planted Solar, Inc. | Robotics Control Engineer — San Francisco Bay Area, CA

Behavior requirements and motion controls design for self-driving vehicle architecture — *July 2026 - Present*

- Built full-stack **ROS2** control simulation integrating torque control, behavior trees, and fault-diagnosis in **Groot2's blackboard**
- Converted 89-state hydraulic steel-post installer **FSM** to **behavior tree** in **C++**, validated with **128 tests** & **Groot2** monitor
- Conducted HIL testing to validate the production-scale behavior of the robot for accurate installation of steel posts on field

Tesla, Inc. | Software Intern - Vehicle Controls — Palo Alto, California

Grey box model of Tesla Model 3 semi-active hydraulic suspension — *June 2025 - September 2025*

Hydraulic dampers exhibit extremely nonlinear behavior due to fluid cavitation during high-current and high-speed stroke operation

- Automated static map generation process using **least-squares** optimization, achieving **78%** fit against ground-truth data
- Cascaded static maps with **globally optimized** exponential smoothing filter to fit force-velocity hysteresis with **83%** accuracy
- Optimized the adaptive smoothing constant as nonlinear function of damper inputs to model cavitation with **92.8%** accuracy
- Built a **hybrid physics-deep learning model** integrating static maps, filtering and neural network, reaching **96%** accuracy
- Delivered **ML-based system identification python tool**, streamlining damper's model development and control design

UCLA Visual Machines Group | Research Assistant — Los Angeles, CA

Physics-benchmarking of World Foundation Models | **Project Website** | **arXiv** — *April 2025 - Present*

World foundation models like Cosmos claim physical accuracy in video prediction - necessitating rigorous scientific verification

- Built **object segmentation pipeline** to predict object kinematics from experimental data on projectile motion and Stoke's law
- Estimated gravity and fluid viscosity from Cosmos' video predictions to benchmark against values computed from experiments

Jaguar Land Rover Ltd. | Motion Controls Engineer — Bangalore, India

Active Ride Functionality | *Dept. of Chassis & Motion Controls Systems* | **Publication** — *July 2021 - August 2024*

Active ride functionality is a vehicle ride enhancement algorithm designed to deliver superior comfort than JLR's adaptive dampers

- Spearheaded the **model-based design of optimal controllers** to reduce road-induced vibrations in Range Rover by **38.6%**
- Authored system requirements for actuators, translating ride KPIs and functional safety goals into verifiable control specification
- Designed **Kalman Filter for state-estimation** of heave velocity, pitch, and roll angle signals with **max RMS error = 0.11**
- Tuned **LQR, MPC**, $\mathbf{H}_\infty$ to limit actuation power consumption to **0.08% of battery capacity** for 30-min-long **WLTP test cycle**
- Implemented **LQR**, **MPC** in **safety-critical C++** and conducted **unit testing** to verify stability, boundedness and robustness

Torque Split for Efficiency (TSE) | *Dept. of Powertrain Energy & Thermal Management Systems* *April 2023 - Sept 2023*

TSE is an optimization algorithm for the most energy-efficient distribution of driver's torque demand between front and rear EDU

- Modified TSE for improved efficiency by integrating **energy consumption maps** for vanes, fans, & pumps in the cost function
- Reported **80Wh** energy saving using the updated algorithm by co-simulating 30min WLTP test cycle in GTSuite + Simulink

Crash Pulse Prediction | *Dept. of Structural Integration* *August 2021 - July 2022*

- Developed a **physics engine** to simulate vehicle crash kinematics with **92% accuracy** in **4 minutes** compared to **8-hour FEA**
- Packaged the engine as a **MATLAB application** to save **20% workdays** in CAD model development of **Panthera**'s body-in-white

AI & Robotics Technology Park | Robot Controls Intern | IEEE CDC'23 Publication

Bangalore, India

Formation control of autonomous mobile robots with input saturation and constraints on formation size *May 2021 - August 2021*

- Developed a novel path-following controller ensuring **99.2% tracking accuracy** and smooth saturation of robot's speed limits
- Proved the **Lyapunov stability** of the controller and demonstrated its scalability for formation control in **ROS2/Gazebo**

University of Toronto | Mitacs Globalink Research Intern | Report

Toronto, Canada

Seismic response of vertical post-tensioned connection for modular steel buildings (MSBs) *April 2020 - August 2020*

- Modeled and simulated the novel vertical post-tensioned connection using **Finite Element Analysis** in **ABAQUS** for assessing the self-centering properties and **30% higher energy dissipation capabilities** compared to welded steel structures
- Reported hysteresis and seismic capabilities of vertical post-tensioned connection with **91% accuracy** relative to experiments
- Researched contact models and fracture criteria in **ABAQUS** to achieve **99.75% accuracy** in the frictional dissipation model

Robotics and Autonomy Projects at UCLA and IIT Bombay

Risk-aware highway lane-change planning using CVaR-MPPI | Report | Code

Fall 2025

- Augmented Conditional Value at Risk into MPPI controller to improve vehicle safety and path tracking during lane-change
- Benchmarked CVaR-MPPI against MPPI on HighwayEnv simulator on three scenarios: hard-brake, cut-in, and lane-change
- Achieved 98.6% increase in **minimum-time-to-collision as per ISO 17387** and 71.2% **improved path-tracking** accuracy
- Evaluated the controller's efficacy on **efficiency** (merge duration), and **comfort** (ISO11270 - max lateral acceleration & jerk)

Reinforcement learning based safe autonomous parking | Report | Code

Fall 2025

- Trained an **RL control policy** using **SAC + HER framework** for reverse and parallel parking in obstacle-rich environment
- Achieved 100% success rate with **curriculum training** across increasing parking difficulty (static –> dynamic obstacles)
- Implemented **MPC** for **motion planning** using a 20K-parameter **neural bicycle behavior model** trained on state-action data

Optimization-based collision avoidance (ObCA) for motion planning | Report | Code

Spring 2025

- Collaborated with a 3-member team to **integrate ObCA within MPC trajectory planner** for collision-free parking maneuvers
- Co-designed and validated cost functions to generate smooth, jerk-free vehicle trajectories in dynamic obstacle environments

Control of 3DoF Robotic Manipulator for Lego Assembly | Report | Code

Spring 2025

- Designed **joint-space inverse dynamics controller** for picking, placeing & fitting Lego blocks with 1.5° **tracking precision**
- Implemented the controller in real-time on a 3D printed 2R-1P manipulator actuated by Dynamixel MX-28 dual shaft servo

Predicting Keystrokes from Electromyography (EMG) Signals | Report | Code

Winter 2025

Training a neural network to predict the keyboard keystrokes from the EMG signals recorded using neural bands on a person's wrists

- Proposed & tested **8 DNN architectures** based on inductive bias of transformers, RNNs, and CNN for the *emg2qwerty* dataset
- Achieved **22% reduction in error** by training Transformers+GRU+CNN architecure over baseline architecture for 50 epochs

MAPLE based solver for stress analysis of a loaded elastic cavity | Report

May 2019 - July 2019

- Engineered a **MAPLE solver** using **general Michell solution** for 2D infinite annular domain with generalized internal traction
- Validated MAPLE results through ABAQUS simulations, modeling continuous and point load distributions in symmetric systems
- Verified **Saint Venant's Principle** for equivalent load distributions by analyzing stress patterns via **MATLAB** visualizations

Image Creating Robotic Arm | Certificate | Report

May 2018 - July 2018

- Designed a **2-link manipulator** in **SolidWorks** and determined the requisite torque capacity for servo motor using **FEA**
- Deployed **Canny Edge Detection** algorithm to convert an input RGB image to **bit-Matrix** for planning end-effector's path
- Determined the joint space trajectory using **Inverse Dynamics** and programmed **Arduino** with **PD control** for servo control